

SAT Practice Module: Gemini Generated

Date: March 7, 2026 **Topic Focus:** Weakness Blueprint (Discriminants, Vertex Form, Unit Conversions, Function Mapping, Logical Conclusions, Grammar Boundaries, Subject-Verb Agreement, Rhetorical Synthesis)

Math (Module 2 Hard Style)

1. The equation $kx^2 + 10x + k = 0$ has no real solutions, where k is a constant. Which of the following best describes all possible values of k ?
- A. $k > 5$
 - B. $k < -5$
 - C. $k > 5$ or $k < -5$
 - D. $-5 < k < 5$
2. A retail company estimates that its monthly profit, $P(x)$, in thousands of dollars, is defined by a quadratic function of x , the amount spent on digital advertising in thousands of dollars. The profit reaches a maximum of 120 thousand dollars when the company spends 8 thousand dollars on advertising. If the company spends 0 dollars on advertising, the profit is -8 thousand dollars. Which equation represents this situation?
- A. $P(x) = -2(x - 8)^2 + 120$
 - B. $P(x) = -2(x + 8)^2 + 120$
 - C. $P(x) = -16(x - 8)^2 + 120$
 - D. $P(x) = -(x - 8)^2 + 120$
3. *Student-Produced Response* An industrial 3D printer builds a model with a volume of 450 cubic inches. The material used for the model has a density of 0.05 kilograms per cubic centimeter. If 1 inch is approximately 2.54 centimeters, what is the mass of the model in kilograms? (Round your answer to the nearest whole number.)
4. The function f is defined by $f(x) = x^2 - 14x + 55$. The function g is defined by $g(x) = f(x + 3)$. What is the minimum value of $g(x)$?
- A. 6
 - B. 7
 - C. 10
 - D. 55
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Reading & Writing (Module 2 Hard Style)

5. *Ecological Niche Modeling* is a technique used by conservationists to predict where species might live based on environmental conditions. In a recent study, researchers modeled the potential range of the

Himalayan Monal, a high-altitude pheasant. The model, based solely on temperature and precipitation data, predicted the bird could inhabit several mountain ranges where it has never been sighted. The researchers hypothesize that the bird's absence from these theoretically suitable habitats is due to biological factors rather than climatic ones.

Which finding, if true, would most directly support the researchers' hypothesis?

- A. The *Himalayan Monal* is found in some areas with temperatures significantly lower than those used in the model.
 - B. The mountain ranges where the bird is absent are heavily populated by a predator that feeds exclusively on the *Himalayan Monal*.
 - C. The vegetation in the mountain ranges where the bird is absent is nearly identical to the vegetation in its known habitat.
 - D. The *Himalayan Monal* has been observed migrating between mountain ranges that are separated by large valleys.
6. The works of contemporary artist Kehinde Wiley, known for his vibrant, naturalistic portraits of Black people in heroic poses, _____ distinct from the traditional European portraiture that inspires them; Wiley replaces the historical subjects with modern-day individuals in street wear.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. is
 - B. are
 - C. has been
 - D. was
7. In 1968, the sheer weight of the Saturn V rocket, combined with the immense thrust required to escape Earth's gravity, presented a unique engineering challenge _____ the rocket's internal guidance computers had to be rugged enough to withstand intense vibrations while maintaining delicate calculations.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. challenge; however
 - B. challenge,
 - C. challenge;
 - D. challenge, consequently
8. While researching a topic, a student has taken the following notes: * *Semiotics* is the study of signs and symbols and their use or interpretation. * Ferdinand de Saussure (1857–1913) was a Swiss linguist. * He is considered a founder of modern semiotics. * He argued that a "sign" consists of two parts: the *signifier* (the form the sign takes) and the *signified* (the concept it represents). * The relationship between the signifier and the signified is arbitrary.

The student wants to explain Saussure's specific definition of a "sign." Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. Considered a founder of modern semiotics, Swiss linguist Ferdinand de Saussure (1857–1913) argued that the relationship between a sign's components is arbitrary.

- B. Ferdinand de Saussure, a Swiss linguist, contributed significantly to the field of semiotics, which is the study of signs and symbols.
- C. According to Saussure, a "sign" is composed of a *signifier* (the form) and a *signified* (the concept), which share an arbitrary relationship.
- D. Saussure defined the *signifier* as the form a sign takes, distinguishing it from the *signified* in his studies of semiotics.